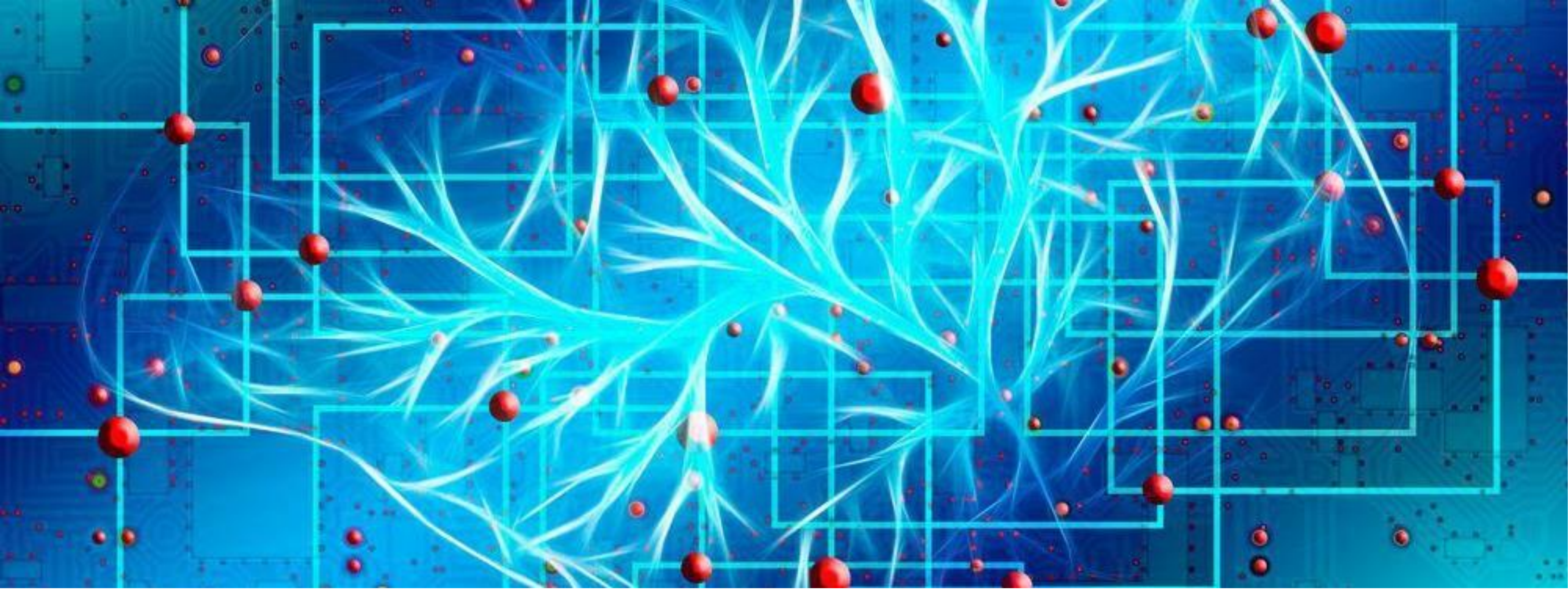


WORD EMBEDDINGS/RNN

Flora Gashi

PREPROCESSING

- 1. Lowercased all text
- 2. Whitespace tokenization
- 3. Removed punctuation (except apostrophes to preserve contractions)
- 4. No stopword removal
- 5. Word embeddings from pretrained FastText (300d)
- 6. Fallback: zero vector for unknown/empty sequences

INPUT/OUTPUT FORMAT: **MLP** (CLASSIFIER)

- **Input**: For each question, create 5 (question, answer) pairs
  - Tokenize both
  - Get FastText embeddings (300d per word)
  - Apply **mean pooling** separately to q and a -> (300 + 300 = **600d**)
  - Shape: (batch_size, 5, 600)

- **Output**:
  - MLP processes each pair → returns 1 logit per answer
  - shape: (batch_size, 5)
  - Prediction = index of highest logit (argmax)

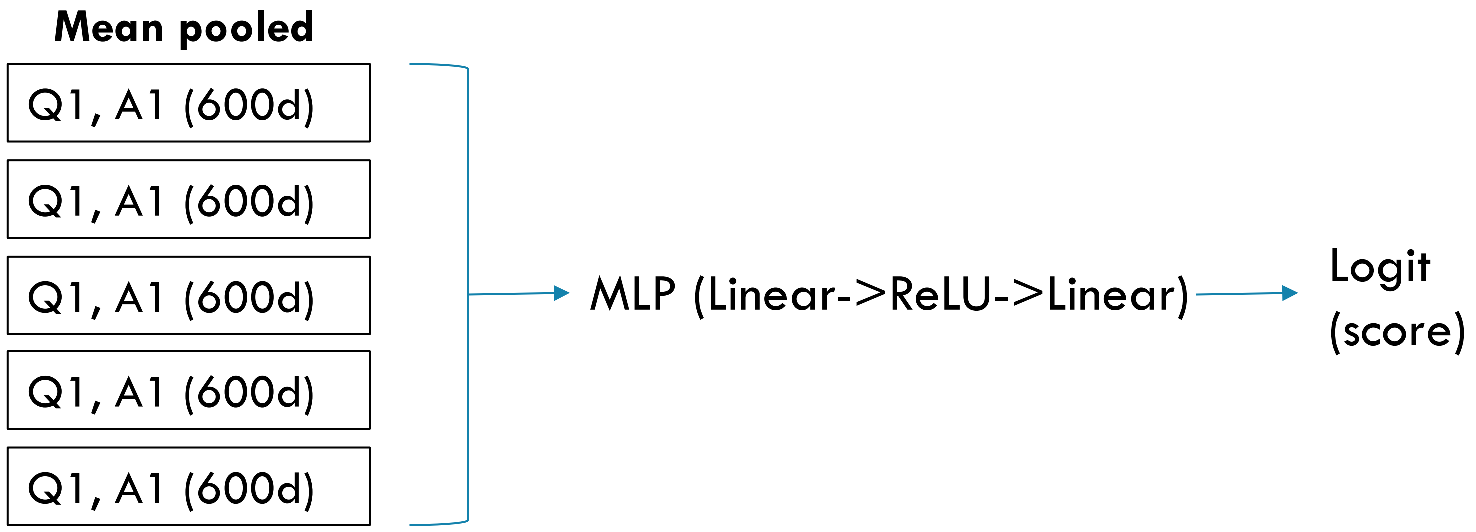
INPUT/OUTPUT FORMAT: **RNN** (GRU)

- **Input**: Combine question + each answer → 5 full sequences
  - E.g “Where do we find magazines? bookstore”
  - Tokenize → FastText embeddings per word (300d)
  - Pad to max sequence length in batch
  - Shape: (batch_size, 5, max_len, 300)

- **Output**:
  - GRU processes each sequence → take final hidden state
  - Classifier returns 1 logit per answer
  - Shape: (batch_size, 5)
  - Prediction = index of highest logit (argmax)

NETWORK ARCHITECTURE: **MLP** (CLASSIFIER)

- One hidden layer with ReLU and dropout
- Each of the 5 question-answer pairs is processed independently
- Outputs a score (logit) for each choice



NETWORK ARCHITECTURE: **RNN** (GRU)

- GRU with 2 layers (uses final hidden state)
- Each of 5 tokenized question-answer pairs is processed separately
- Final classifier layer gives 1 score per option

Q1+A1 tokens → FastText embeddings → GRU → Classifier → logit

Q2+A2 tokens → FastText embeddings → GRU → Classifier → logit

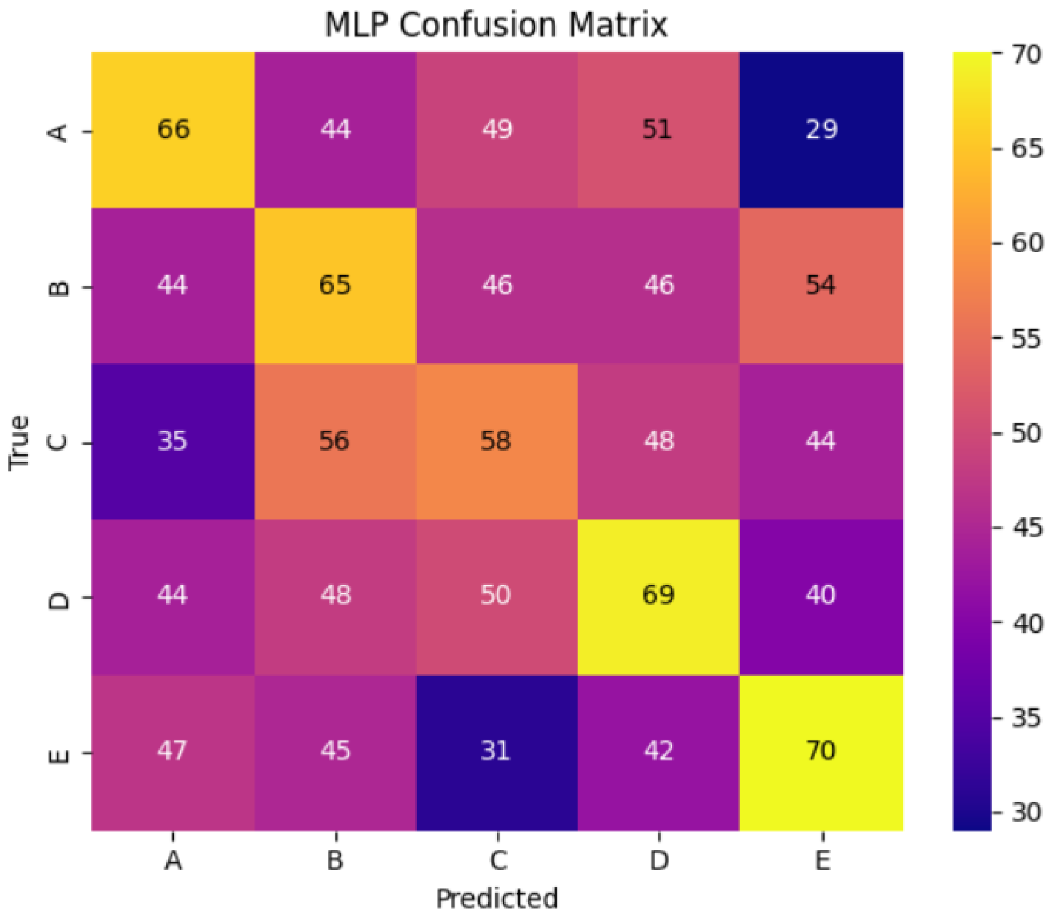
. . . **5x**

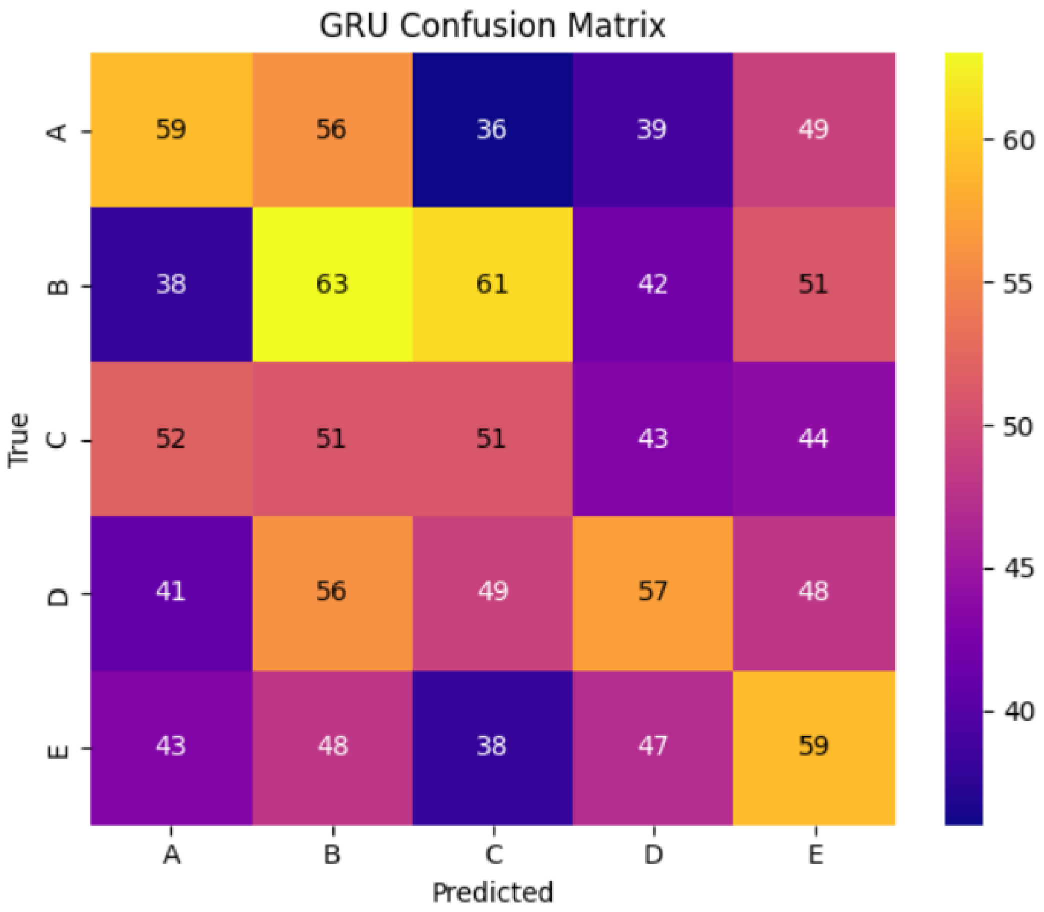
EXPERIMENTS

- Grid search over:
  - Learning Rate: 1e-3, 1e-4, 1e-5
  - Dropout: 0.1, 0.3
  - Hidden dim: 128, 256
- CrossEntropyLoss & Adam optimizer
- Batch size: 128
- 12 runs per model (except GRU missing 1 due to disconnect)
- Early Stopping patience = 12 for 40 epochs
- Best model selected by validation accuracy
- Tracked in WandB: train loss, train accuracy, val loss, val accuracy

RESULTS

Model	Validation Accuracy	Test Accuracy
Baseline	20 %	20 %
MLP	25.6 %	26.9 %
GRU	25.2 %	23.7 %





ERROR ANALYSIS & OBJECTIVES

- Printed 10 misclassified examples/model
- Common error pattern:
  - Confusion between semantically similar distractors
  - E.g. «make peace» vs. «make noise» or «kill animals» vs. «complete job»
- Some plausible even to humans
- Accuracy by question prefix (e.g. «what», «where», «if»)
  - «WHERE» -> highest accuracy
  - «WHEN» -> lowest (especially hard)
- Selected 20 examples where models disagreed

THANK YOU FOR YOUR ATTENTION!